Project Outline

Motivation

Plasma electrolytic oxidation (PEO) grows dense, hard oxide coatings on aluminium through high-voltage electrochemical discharge — producing surfaces with superior wear and corrosion resistance for aerospace and biomedical applications. Despite its industrial importance, the physics governing oxide growth are not well understood at the microstructure scale. Existing models are phenomenological and cannot generalise across processing conditions.

Central question: *Can a fully physics-driven computational framework predict the microstructural and mechanical properties of PEO coatings on aluminium as a function of processing conditions?*

Research Roadmap

1
Fracture Model
CP + phase-field fracture
(this poster)

→

2
Electrolyte–Interface Breakdown Model
high-field ion transport

→

3
Oxide Growth Model
interface kinetics
+ microstructure evolution

→

4
Fully Coupled PEO Growth Model
processing → properties

Phase 1 — Current Work (this poster)

An Eulerian crystal plasticity framework (`CPFoam`) is being extended to incorporate phase-field fracture, treating crack propagation as an additional order-parameter field φ_{cr} evolving under a free-energy competition between elastic strain energy W_{el} and grain-boundary interfacial energy W_{int}. The two-grain bicrystal results on the poster demonstrate:

- Grain-boundary-directed crack propagation under mode I dynamic loading
- Correct energy competition: W_{int} localises at the GB; W_{el} drives nucleation
- Identification of key numerical challenges in the fully coupled regime

Outlook

The ultimate goal is a modular OpenFOAM toolkit capable of taking PEO processing conditions (voltage regime, electrolyte composition, temperature) as inputs and predicting the resulting coating microstructure and mechanical properties — hardness, residual stress, crack morphology, and oxide thickness — without phenomenological fitting.

- **Phase 2** will model high-field ion transport and dielectric breakdown at the electrolyte–oxide interface, capturing the discharge events that drive oxide growth
- **Phase 3** will couple the breakdown model to oxide growth kinetics, resolving the evolving metal–oxide interface and coating microstructure
- **Phase 4** brings all components together into a fully coupled PEO growth model validated against experimental coating data on aluminium